MACHINE LEARNING AND AI FOR ANALYTICAL CHEMISTRY

Part 3: Evaluation of trained models

M2 Sciences analytiques - 3 h lectures + 1 h tutorial

WHAT THIS PART COVERS

Roadmap for Part 3

1

Failure modes

Bias, overfitting and underfitting, information leakage, and generalisation

2

Control and validation

Dataset roles, cross-validation, complexity and pruning, feature selection, explainable AI

3

Uncertainty

Uncertainty quantification and calibration for defensible decisions

Failure modes

How a model can look good and be wrong

FAILURE MODES

Bias, overfitting, underfitting

Bias

A model reproduces the skew of its training data: most accurate where data are dense, worse elsewhere, while the average still flatters.

Overfitting vs underfitting

Overfitting memorises noise and fails to generalise; underfitting is too simple to capture the pattern. The signature of overfitting is a large train-to-test gap.

Example: a contaminant classifier trained on summer samples performs worse in winter; the headline accuracy hides the gap.

FAILURE MODES

Information leakage (the dangerous one)

Leakage is when information unavailable at prediction time slips into training, inflating measured performance. The model looks excellent until it meets genuinely new data.

Two classic leaks: replicates or adducts of the same molecule on both sides of the split; and scaling or feature selection done on the whole dataset before splitting.

Example: in retention prediction, split by molecule so no compound appears in both train and test; otherwise the model recognises, not predicts.

Generalisation is the goal

Generalisation is performance on data the model has never seen, drawn from the conditions where it will actually be used.

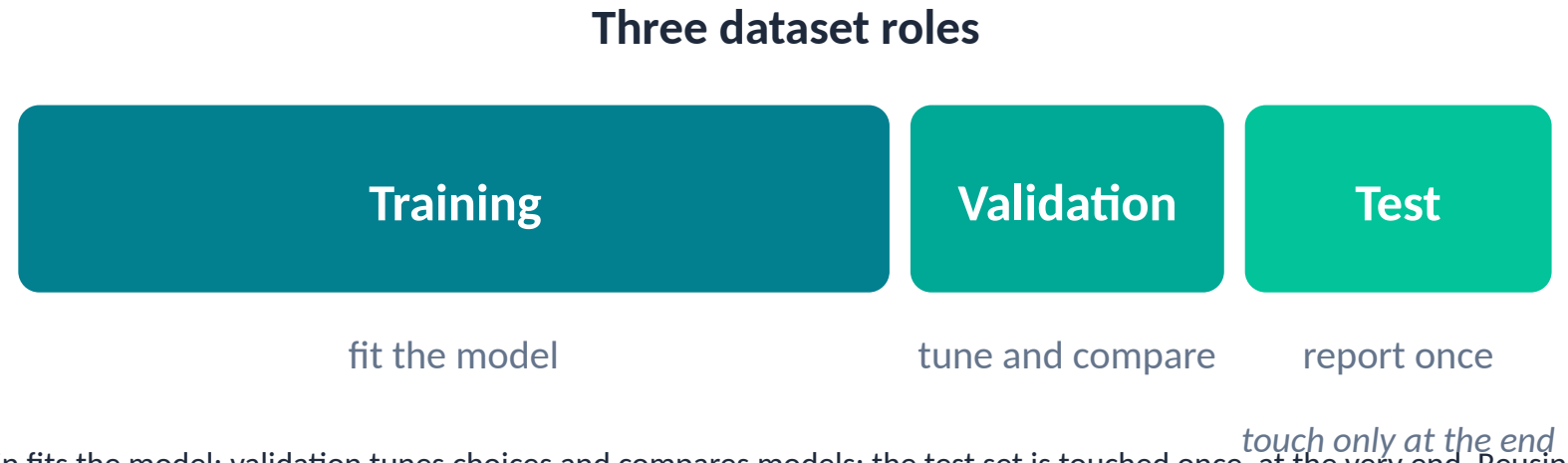
Everything in the rest of this part exists to estimate generalisation honestly, rather than to flatter the model on data it has already met.

Example: the only score that matters is the one on samples measured after the model was built, on a new batch or instrument.

Control and validation

Estimating generalisation, and steering complexity

The three dataset roles



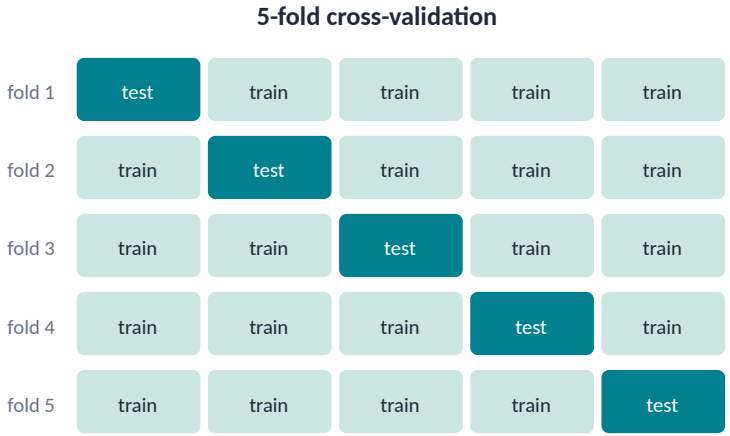
Train fits the model; validation tunes choices and compares models; the test set is touched once, at the very end. Reusing the test set to make choices turns it into a training set and inflates the result.

Cross-validation

When data are scarce, one split is wasteful and noisy.

k-fold rotates the test part so every sample is tested once; average the folds for a stable estimate.

Grouped CV keeps all measurements of one entity in one fold, preventing leakage.



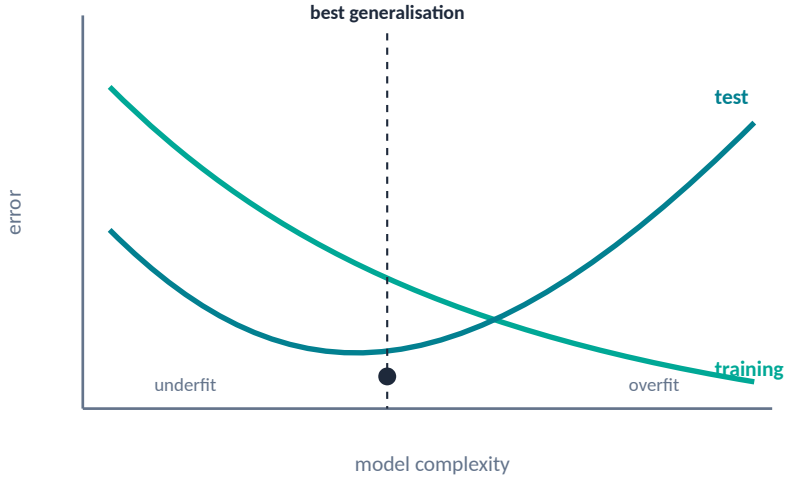
every sample is tested once; average the folds

Example: with 150 spectra, 5-fold CV uses all of them; group by compound if several spectra share a compound.

Model complexity and pruning

Error against complexity is U-shaped: too simple underfits, too complex overfits.

Pruning a tree, limiting depth, or adding **regularisation** moves toward the sweet spot; cross-validation locates it.



Example: cross-validated accuracy peaks at a moderate tree depth, then declines; choose that depth.

Feature selection and analysis

Fewer, well-chosen features often generalise better than many: selection reduces overfitting and improves interpretability.

Crucial caveat: selection must happen inside the cross-validation, on training data only, or it leaks.

Example: picking the ten most discriminating m/z bins helps, but choosing them on the whole dataset before splitting inflates the score.

Explainable AI

An accurate but opaque model is hard to trust in a regulated setting.

Explainable AI (feature importances, SHAP values, partial dependence) describes what drives a prediction, letting a chemist check the model relies on sensible signals.

Example: SHAP showing a toxicity model keys on a known reactive substructure builds trust; keying on a sample-id artefact reveals leakage.

Uncertainty and calibration

A prediction without an uncertainty is of limited use for a decision.

Uncertainty quantification attaches a credible range or probability. **Calibration** checks that stated probabilities match observed frequencies: among 80 %-confidence predictions, about 80 % should be correct.

Example: a calibrated 95 % non-compliance flag supports a defensible decision; an overconfident uncalibrated 99 % invites unjustified action.

KEY TAKEAWAYS

- Good scores on seen data prove nothing; only generalisation counts.
- Leakage is the most common inflator; split by molecule, scale inside the split.
- Use train/validation/test roles; touch the test set once.
- Cross-validation (grouped where needed) gives stable estimates and finds complexity.
- Report uncertainty, and check it is calibrated.